

DCF VALUATION OF COMPANIES

A two-stage template, and building it as a skill or an agent

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Thu Sep 17, 2026

Overview

- 1 Build a template for two-stage DCF
- 2 But firm valuation is not “one size fits all,” so a spreadsheet template or a classical app is limited
- 3 Create an agent skill
- 4 Create a stand-alone agent

Template

INPUT	BASIS
Sales growth	
EBITDA margin	
Depreciation	% of prior year net PP&E
Cash tax rate	cash taxes / pretax income
Sales-to-NWC ratio	prior or average NWC?
Sales-to-net PP&E ratio	prior or average net PP&E?
Cap ex	a plug

Cash taxes: the provision less its deferred portion, or the taxes paid per the cash flow statement.

Issues (There Are Many)

- Value real estate, reserves, ...
- Leases: financial or operating?
- Minority interests, investments in and income from affiliates
- Goodwill: constant or growing?
- Long-term deferred taxes: debt or equity?
- Other long-term assets and liabilities
- Derivative positions
- Preferred stock, convertible bonds, ...

Skills

A folder with a `SKILL.md` in it — written instructions, plus any scripts and examples they refer to.

```
.claude/skills/dcf/  
├── SKILL.md          a description, then the instructions  
├── references/       the template, the issues  
└── scripts/         code the agent can run
```

At startup Claude reads only each skill's description — that is all it holds. When a task matches one, it reads that whole `SKILL.md` and follows it.

The Same File in Other Agents

Anthropic wrote the format and released it as an open standard. The others adopted it rather than inventing their own.

AGENT	WHERE IT LOOKS
Claude Code	<code>.claude/skills/</code>
Codex and ChatGPT	<code>.agents/skills/</code>
Gemini CLI	<code>.gemini/skills/</code> , or <code>.agents/skills/</code>

Same file, same description-first loading. Put the folder in `.agents/skills/` and both Codex and Gemini CLI find it.

Creating a Skill

- Work with AI until you get what you want
- Tell AI to create a skill so you can get the same thing the next time
- Give the skill a meaningful name.
- As you use it and find things to add or modify, tell AI to edit the skill.

What Goes in the DCF Skill

The template lives here.

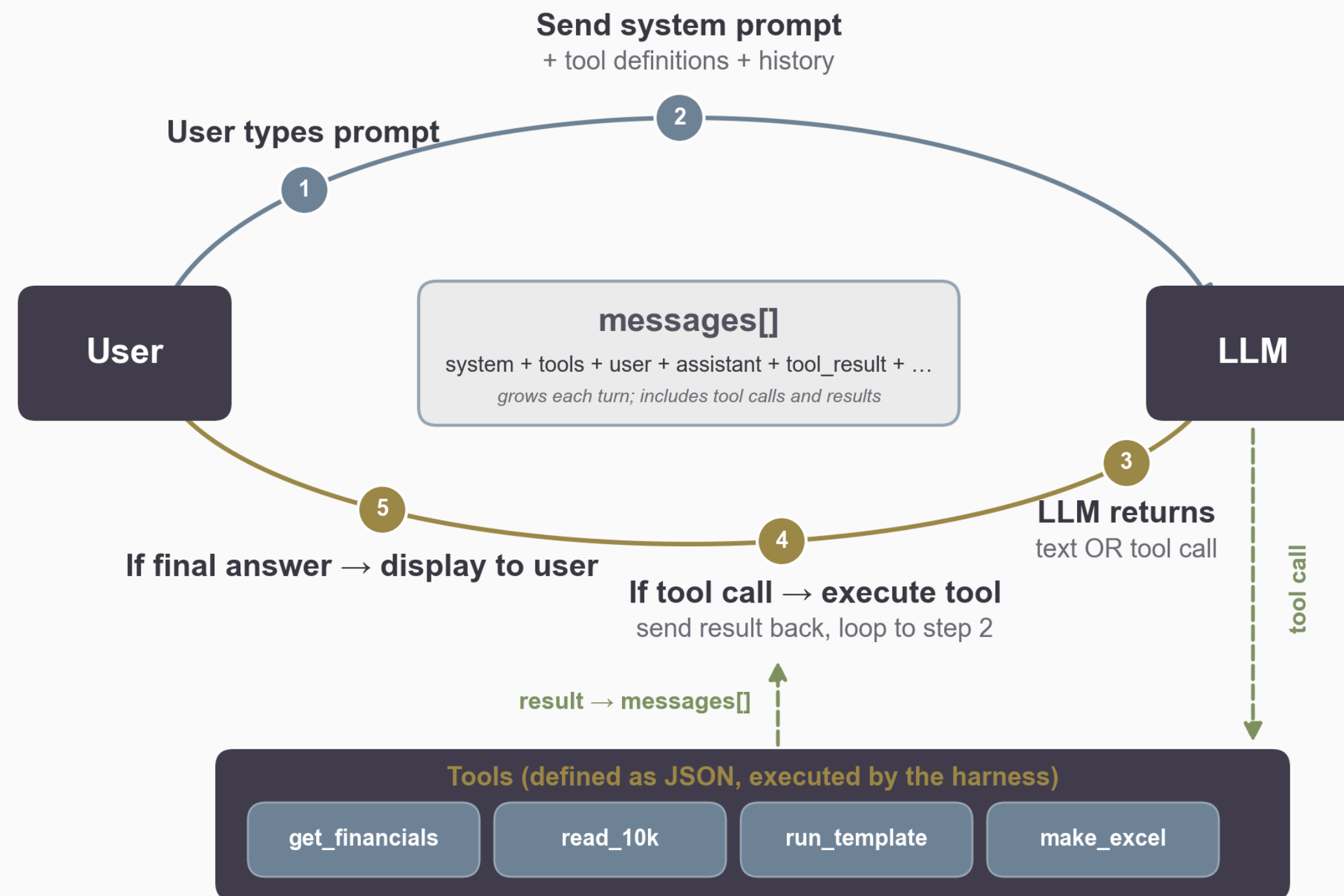
- Instructions for getting data, possibly including MD&A, risks from 10-K, earnings calls, news
- Instructions to use xlsx and pptx skills, any special formatting you want
- Guidelines for walking through other issues

Creating an Agent

- 1 Stand-alone app
- 2 API calls to large language model
- 3 Could use Claude Agent SDK, though Claude can also write everything from scratch

The same `SKILL.md` works in Claude Code and inside the agent app.

Inside an Agent



CLAUDE AGENT SDK

Three Varieties of Tools

Claude Code tools

The inherited built-ins —
Read, Write, Edit, Bash,
Grep, Glob, WebSearch,
WebFetch, ...

Internal MCP tools

Your own Python functions
— `run_python`, or the
valuation itself.

External MCP tools

Connectors — databases,
the web, your inbox.

The LLM never runs anything. The harness — everything in an agent that is not the model — executes the tool the model asks for, or refuses.

Where They Run

Read, Write, Edit, Bash, Grep

the agent's own process — its filesystem, its shell, its network

Internal MCP tools

the same process — whatever it can import, it can run

External MCP tools

someone else's server — arguments out over HTTP, only the result back

WebSearch

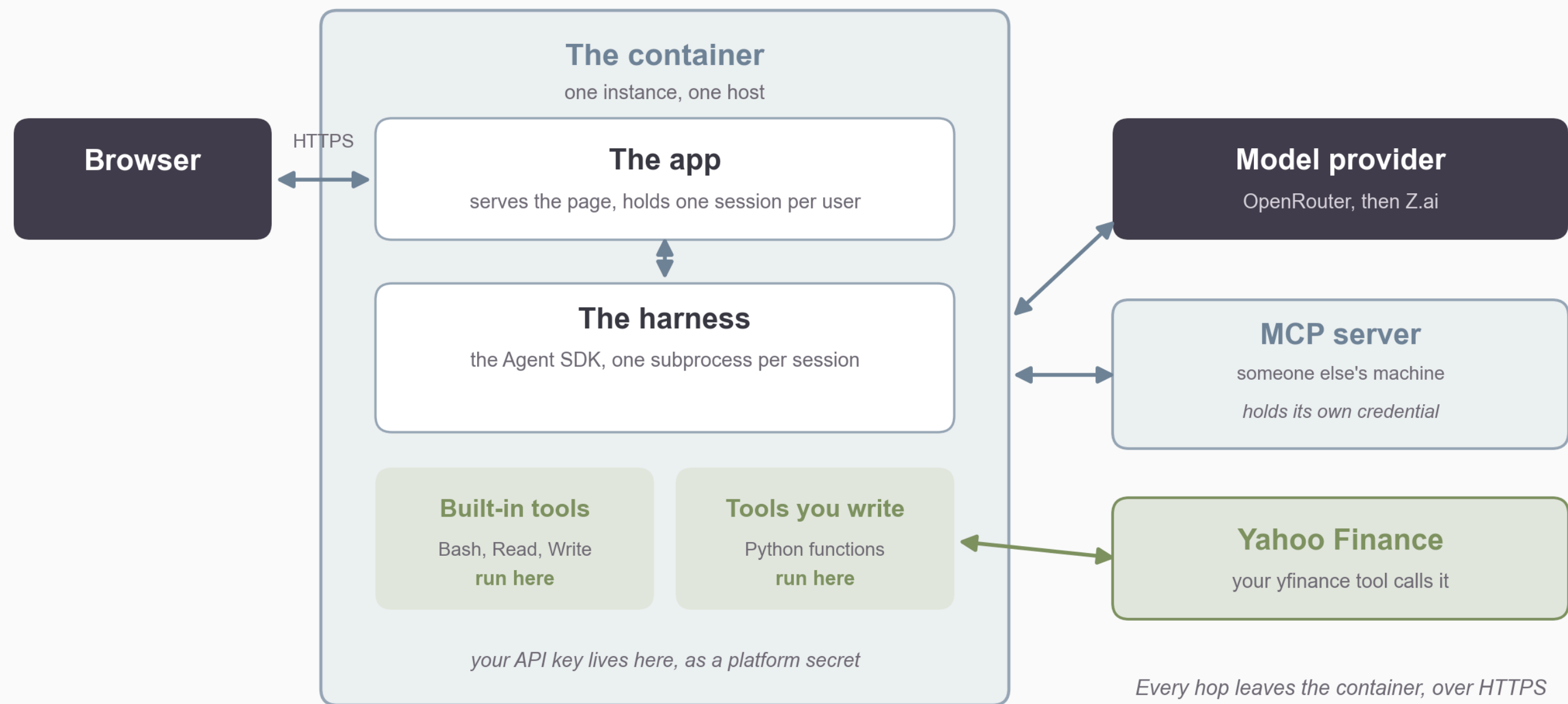
Anthropic's search backend — titles and URLs, not pages

WebFetch

downloads from your machine, then a small model extracts what was asked

So “on your machine” holds for the filesystem and shell tools. The two web tools are not local: the search runs at Anthropic, and the fetched page is sent there to be read.

The Deployed Agent



Agent Options

`ClaudeAgentOptions`

`system_prompt`

a string, or a file on disk when it is long

`tools`

which built-ins to include — `tools=[]` for none

`disallowed_tools`

takes a tool out of the model's context entirely

`allowed_tools`

the tools that run without prompting

`mcp_servers`

the servers to connect, plus the one holding your tools

`max_turns`

how many tool-use round trips before the app stops it

`allowed_tools` is not a whitelist. A tool left off it still runs — it falls through to `permission_mode`.

Permission Modes

`permission_mode`

`default`

prompts before file edits, shell commands, and network requests

`acceptEdits`

auto-approves edits and common filesystem commands

`plan`

read-only — proposes changes but makes none

`dontAsk`

only `allowed_tools` run; anything else is denied, not prompted

`bypassPermissions`

everything runs, no checks — sandboxed containers only

Custom Tools

A Python function with a decorator giving its name, description, and arguments.

```
1 @tool(  
2     "get_financials",  
3     "Fetch a company's income statement, balance sheet and cash flow.",  
4     {"ticker": Annotated[str, "The stock's ticker symbol"]},  
5 )  
6 async def get_financials(args):  
7     t = yf.Ticker(args["ticker"])  
8     csv = t.income_stmt.to_csv()  
9     return {"content": [{"type": "text", "text": csv}]}  
10 local_server = create_sdk_mcp_server(name="local", tools=[get_financials])
```

Tools fetch. The skill specifies the workbook, and the workbook's own formulas do the arithmetic — so the valuation is not a tool.

Producing the Output

The workbook

The `xlsx` skill — pandas and openpyxl, plus a financial-modeling standard for formatting and formula conventions.

The deck

The `pptx` skill — `markdown` and the pack scripts to edit a template, `pptxgenjs` under Node to build one from scratch.

The xlsx skill's rule: formulas in the cells, not values computed in Python. `=SUM(B2:B9)`, never the number Python got — so changing an assumption recalculates the workbook.

Skills are files in `.claude/skills/`, loaded when relevant. The container image has to carry what they import.